

## BARIS ASKIN

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### EDUCATION

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**Carnegie Mellon University, Pittsburgh, USA** *Fall 2022 - Spring 2027 (Anticipated)*  
· Ph.D. in Electrical and Computer Engineering (M.Sc. en route, 2025) **GPA: 4.0/4.0**  
*Advisors: [Gauri Joshi](#) and [Carlee Joe-Wong](#)*

**Bilkent University, Ankara, Turkey** *2017 - 2022*  
· B.Sc. in Electrical and Electronics Engineering **GPA: 3.99/4.0**  
*Valedictorian – Highest Ranked Graduate* of 2022 in the Faculty of Engineering

### RESEARCH AREAS

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LLMs, Interpretability, Alignment, Post-Training & RL (RLVR, distillation, fine-tuning), LLM Routing, Distributed/Federated Learning

### WORK & RESEARCH EXPERIENCE

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**Graduate Researcher** *August 2022 – Present*  
*Carnegie Mellon University*

- LLM Alignment: emergent & subliminal misalignment analysis through training data and signal (SFT, on-/off-policy distillation) [[NeurIPS'26](#)]
- LLM Interpretability: info.-theoretic method for internal planning horizon and branch awareness [[ICLR'26](#)]
- LLM RLVR: federated GRPO with off-policy coordination on public data [[ICML'26 Workshop](#)]
- LLM Routing: learning routers from sparse, decentralized evaluations in federated settings [[NeurIPS'26](#)]
- Federated Learning and Fine-Tuning: communication-efficient fine-tuning (LoRA) and asynchronous FL with convergence guarantees [[NeurIPS'25](#)][[AISTATS'25](#)][[UAI'24](#)]

**Research Intern** *May 2025 – August 2025*  
*NVIDIA Corporation*

- Developed asynchronous FL with data-free knowledge distillation to mitigate staleness [[NeurIPS'26](#)]

**Quant Researcher Intern** *June 2026 – August 2026*  
*IMC Trading*

- Quantitative research on index options.

**Undergraduate Researcher** *September 2020 – July 2022*  
*Imaging and Computational Neuroscience Lab, Bilkent University, Ankara, Turkey*

- Developed deep learning methods (transformers, CNNs, deep equilibrium models) for medical imaging.
- Resulted in two IEEE Trans. on Medical Imaging papers and four other conference/journal papers.

### SELECTED PUBLICATIONS

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- Emergent and Subliminal Misalignment Through the Lens of Data-Mediated Transfer**  
**B. Askin\***, M. Ustaomeroglu\*, A. Nayak\*, G. Joshi, G. Qu, C. Joe-Wong  
*NeurIPS 2026* [[Link](#)]
- Internal Planning in Language Models: Characterizing Horizon and Branch Awareness**  
M. Ustaomeroglu\*, **B. Askin\***, G. Joshi, C. Joe-Wong, G. Qu *\*equal contribution*  
*ICLR 2026* [[Link](#)][[Poster](#)]

3. **Reviving Stale Updates: Data-Free Knowledge Distillation for Asynchronous FL**  
B. Askin, H. R. Roth, Z. Sun, C. Joe-Wong, G. Joshi, Z. Xu  
*NeurIPS 2026* [[Link](#)][[Code](#)]
4. **Federate the Router: Learning LM Routers with Sparse and Decentralized Evaluations**  
B. Askin\*, S. Patel\*, A. Nayak\*, A. Vigano, J. Woo, G. Joshi, C. Joe-Wong  
*NeurIPS 2026* [[Link](#)][[Code](#)]
5. **PubSwap: Public-Data Off-Policy Coordination for Federated RLVR**  
A. Nayak\*, B. Askin\*, M. Ustaomeroglu, C. Joe-Wong, G. Joshi  
*ICML 2026 Workshop on RL from World Feedback* [[Link](#)]
6. **Federated Communication-Efficient Multi-Objective Optimization**  
B. Askin, P. Sharma, G. Joshi, C. Joe-Wong  
*AISTATS 2025* [[Link](#)] [[Code](#)][[Poster](#)]
7. **Ravan: Multi-Head Low-Rank Adaptation for Federated Fine-Tuning**  
A. Raje, B. Askin, D. Jhunjhunwala, G. Joshi  
*NeurIPS 2025* [[Link](#)]
8. **FedAST: Federated Asynchronous Simultaneous Training**  
B. Askin, P. Sharma, C. Joe-Wong, G. Joshi  
*UAI 2024* [[Link](#)][[Code](#)][[Poster](#)]
9. **Federated Learning is a Lens towards a Democratized Future for the Scaling Law Era**  
H. H. Jiang, B. Askin, G. Joshi, C. Joe-Wong  
*ICML 2026 (Position Track)* [[Link](#)]

For the complete list, including my earlier deep learning for medical imaging work published in IEEE journals and conferences, please see my [Google Scholar](#).

## SELECTED COURSES AND RELEVANT SKILLS

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**CMU:** Ph.D.-level Optimization and Statistics courses, Generative AI, MDPs and RL, Deep RL

**Bilkent:** Linear Algebra, Stochastic Models, Signals and Systems, Digital Signal Processing

**Skills:** Python, PyTorch, Hugging Face Libraries, vLLM, VERL, OpenCV, Apache Spark

## PROFESSIONAL SERVICE

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- Reviewer for NeurIPS, ICLR, ICML, AISTATS, ISIT, AAAI, TMLR, IEEE/ACM ToN and MLSys
- Teaching Assistant for [Introduction to ML for Engineers \(18-661\)](#) and [Algorithms for Large-scale Distributed ML and Optimization \(18-667\)](#)

## SELECTED AWARDS AND ADDITIONAL INFORMATION

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- 2023-2024 Ben Cook Presidential Graduate Fellowship at CMU
- 2022-2023 Carnegie Institute of Technology Dean's Fellowship at CMU
- Bilkent University Comprehensive Scholarship: Full tuition waiver & stipend during the B.Sc.
- Scholarship of Turkish Ministry of Youth and Sports: Awarded stipend during the B.Sc.
- Ranked 252nd/2M in the university and 3rd/104k in the postgraduate entrance exams in Turkey
- Volunteer at Young Guru Academy (an international NGO based in Istanbul) from 2018 to 2020